

Edge-Efficient Knowledge-Distilled Hybrid Vision Model for Plant Disease Detection

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Abstract. Plant disease detection plays a crucial role in precision agriculture and food security. However, deploying highly accurate deep learning models on edge devices remains challenging due to computational and memory constraints. In this work, we propose an edge-efficient hybrid vision framework that leverages a ConvNeXt V2-inspired teacher model to distill knowledge into a lightweight MobileNetV2 student architecture. The proposed teacher incorporates large-kernel depthwise convolution, Layer Normalization, and Global Response Normalization (GRN) to learn rich visual representations from plant leaf images. Knowledge is transferred to the student through a dual-loss knowledge distillation strategy combining hard-label cross-entropy loss and temperature-scaled Kullback–Leibler divergence. Experimental evaluation demonstrates that the lightweight MobileNetV2 student maintains competitive classification performance while substantially reducing model complexity. The final student contains 2.23 million parameters compared with 27.82 million parameters in the teacher, corresponding to approximately 92% parameter reduction. Furthermore, a Generative AI module implemented using Ollama is integrated after classification to provide contextual agricultural recommendations. The resulting framework combines plant disease classification, computational efficiency, and intelligent agricultural assistance, making it suitable for resource-constrained edge deployment.

Keywords: Plant Disease Detection · Knowledge Distillation · MobileNetV2 · ConvNeXt V2 · Edge AI · Lightweight Deep Learning · Generative AI

1 Introduction

Plant diseases significantly affect agricultural productivity and can result in substantial economic losses when they are not detected at an early stage. Conventional disease identification generally depends on visual inspection by farmers or agricultural experts. Although such approaches can be effective, they are time-consuming, subjective, and difficult to scale across large agricultural regions.

Deep learning-based image classification has emerged as an effective approach for automatic plant disease recognition. Convolutional neural networks can learn discriminative visual characteristics directly from images and have demonstrated strong performance across a wide range of visual recognition tasks. Representative architectures include ResNet [1], DenseNet [2], EfficientNet [3], and VGG [4].

Modern convolutional architectures have increasingly focused on improving both representation quality and computational efficiency. ConvNeXt demonstrated that modernized convolutional networks can achieve competitive performance with contemporary vision architectures [5]. ConvNeXt V2 subsequently introduced a redesigned ConvNeXt family together with Fully Convolutional Masked Autoencoders and Global Response Normalization (GRN) [6].

Despite their accuracy, high-capacity deep learning models can be difficult to deploy on resource-constrained devices. Agricultural applications frequently require inference on mobile devices, Internet of Things (IoT) systems, embedded computers, and other edge platforms where memory, computational power, and energy availability are limited.

Lightweight architectures such as MobileNet and MobileNetV2 were specifically designed to reduce computational requirements while retaining strong visual representation capability [7,8]. MobileNetV2 in particular uses inverted residual blocks and linear bottlenecks to obtain a favorable balance between accuracy and computational efficiency [8].

Knowledge distillation provides another mechanism for reducing deployment complexity. Instead of deploying a large teacher network directly, a smaller student network can learn from both ground-truth labels and the soft predictions produced by a high-capacity teacher [9]. The softened probability distribution contains additional information about relationships between classes that may not be represented by one-hot ground-truth labels.

In this work, we propose a hybrid teacher–student framework for plant disease detection. A ConvNeXt V2-inspired architecture is used as the teacher, while MobileNetV2 is used as the lightweight student. The teacher incorporates large-kernel depthwise convolution, Layer Normalization, channel-wise processing, and Global Response Normalization (GRN) to obtain rich visual representations. The student is trained using both ground-truth labels and softened predictions generated by the teacher.

The main contributions of this work are as follows:

- A ConvNeXt V2-inspired teacher architecture incorporating Layer Normalization and Global Response Normalization for plant disease representation learning [6].
- A lightweight MobileNetV2 student architecture designed for computationally efficient edge inference [8].
- A dual-loss knowledge distillation strategy combining hard-label classification loss and temperature-scaled soft distillation loss [9].
- A comparative evaluation of several transfer learning architectures with respect to accuracy, parameter count, computational complexity, and training time.
- Integration of a Generative AI advisory component after disease classification to provide contextual agricultural recommendations.

The remainder of this paper is organized as follows. Section 2 reviews related work and identifies the research gap. Section 3 describes the dataset and task

formulation. Section 4 presents the proposed methodology. Section 5 discusses the experimental evaluation and results. Section 6 presents the discussion and limitations. Section 7 concludes the paper and presents future research directions.

2 Related Work

This section reviews prior work relevant to lightweight image classification, modern convolutional architectures, and knowledge distillation. The discussion focuses on the architectural choices motivating the proposed teacher-student framework.

2.1 Deep Convolutional Architectures

Residual learning was introduced to enable the effective optimization of deeper convolutional neural networks through residual connections [1]. ResNet became a widely adopted backbone for image classification and transfer learning.

DenseNet subsequently introduced dense connectivity between layers, allowing feature reuse and improving information flow throughout the network [2]. EfficientNet proposed compound scaling of network depth, width, and input resolution to obtain improved accuracy-efficiency trade-offs [3].

VGG demonstrated that increasing network depth using small convolutional filters can produce strong visual representations [4]. However, the resulting models can contain a large number of parameters, making them less attractive for resource-constrained deployment.

These architectures establish the importance of balancing representation capacity with computational complexity when designing image classification systems.

2.2 Lightweight Vision Models

MobileNet introduced depthwise separable convolutions as a mechanism for reducing the computational cost of conventional convolutional operations [7]. MobileNetV2 further improved efficiency through inverted residual blocks and linear bottlenecks [8].

The resulting architecture provides a strong basis for edge-oriented computer vision systems because the model can achieve competitive classification performance with substantially fewer parameters and operations than many conventional CNN architectures.

For the proposed framework, MobileNetV2 is therefore selected as the student network. Its compact architecture makes it suitable for deployment after knowledge distillation.

2.3 ConvNeXt and ConvNeXt V2

ConvNeXt revisited conventional convolutional networks using modern architectural design principles and demonstrated that convolutional architectures can remain highly competitive for visual recognition [5].

ConvNeXt V2 further developed this direction by introducing Fully Convolutional Masked Autoencoders and Global Response Normalization [6]. GRN is intended to improve feature diversity and channel-wise representation.

The proposed teacher architecture is inspired by these design principles. Rather than directly deploying a full ConvNeXt V2 model, we use a ConvNeXt V2-inspired teacher incorporating large-kernel depthwise convolution, Layer Normalization, channel-wise processing, and GRN.

This design allows the teacher to learn rich representations that can subsequently be transferred to the lightweight student.

2.4 Knowledge Distillation

Knowledge distillation was introduced as a method for transferring knowledge from a large model to a smaller model [9]. The student learns not only from hard ground-truth labels but also from the probability distribution generated by the teacher.

Temperature scaling is commonly used to soften the teacher distribution, allowing the student to learn relative relationships between different output classes [9].

The approach has subsequently been extended to multiple architectures and tasks, including intermediate-representation transfer and attention-based distillation [10,11]. Such approaches demonstrate the usefulness of teacher-generated information when training compact models.

In the proposed work, knowledge distillation is used to transfer information from the ConvNeXt V2-inspired teacher to MobileNetV2. The teacher is used only during training, while the student is retained for inference.

2.5 Research Gap

Existing CNN architectures provide strong image classification performance, but many high-capacity architectures remain computationally expensive for edge deployment [1–6]. Lightweight architectures such as MobileNetV2 address this issue but generally operate with substantially lower representational capacity [8].

Knowledge distillation provides a mechanism for transferring information from a high-capacity model to a compact model [9–11]. However, a practical plant disease detection framework must simultaneously consider classification performance, model size, computational complexity, and deployment efficiency.

The research gap addressed in this work is therefore the development of a hybrid teacher–student framework in which a high-capacity ConvNeXt V2-inspired model provides training supervision to a lightweight MobileNetV2 student for plant disease classification.

Table 1. Dataset Summary

Attribute	Value
Image Resolution	224×224
Color Format	RGB
Number of Classes	3
Training Images	1322
Testing Images	150
Task Type	Multi-class Classification

The proposed approach combines modern convolutional representation learning, knowledge distillation, and lightweight inference into a unified pipeline.

3 Dataset Description

3.1 Data Source and Task Formulation

This study utilizes the publicly available Plant Disease Recognition Dataset used for supervised plant disease classification.

The images contain variations in lighting, orientation, background, and visual appearance. Such variations provide a suitable environment for evaluating the ability of deep learning models to learn disease-related visual features.

The problem is formulated as a supervised multi-class image classification task. Given an input image

$$x \in \mathbb{R}^{224 \times 224 \times 3}, \quad (1)$$

the objective is to predict the corresponding disease label

$$y \in \{1, 2, \dots, C\}, \quad (2)$$

where C denotes the number of disease categories.

The model learns a parametric mapping:

$$f_{\theta}(x) = \hat{y}, \quad (3)$$

where θ represents the learnable parameters and $\hat{y}$ represents the predicted probability distribution over the disease classes.

All images are resized to 224×224 pixels before being provided to the models. Standard tensor normalization and preprocessing transformations are applied during training.

3.2 Dataset Statistics

The dataset configuration used in this work is summarized in Table 1.

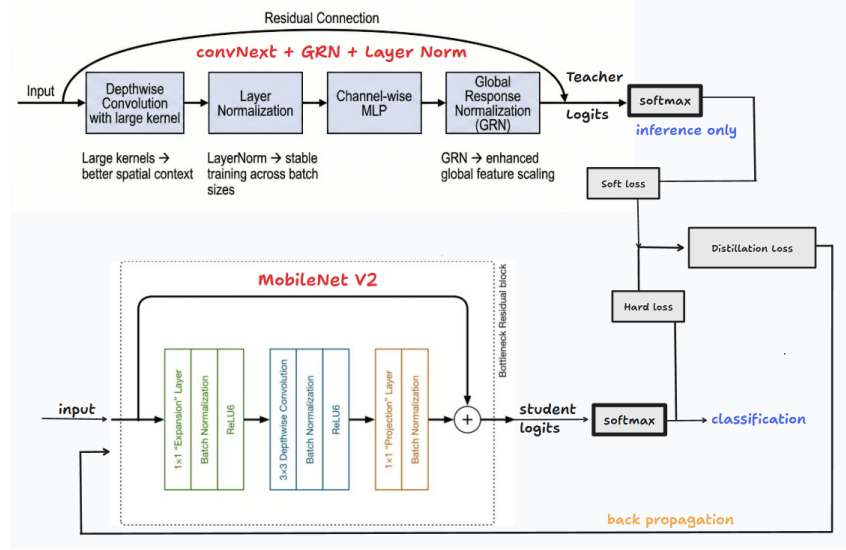


Fig. 1. Proposed hybrid knowledge distillation framework. The ConvNeXt V2-inspired teacher employs large-kernel depthwise convolution, Layer Normalization, channel-wise processing, and Global Response Normalization (GRN). The teacher produces soft logits that supervise the MobileNetV2 student during training. The student additionally receives hard-label supervision through classification loss. During inference, the teacher is removed and only the lightweight student is used for plant disease classification.

4 Proposed Methodology

The proposed framework consists of four major components: (i) a ConvNeXt V2-inspired teacher network, (ii) a MobileNetV2 student network, (iii) a dual-loss knowledge distillation mechanism, and (iv) a Generative AI advisory module.

The overall architecture is presented in Fig. 1.

As illustrated in Fig. 1, the proposed system follows a teacher–student knowledge distillation pipeline designed for edge-efficient plant disease detection.

A high-capacity ConvNeXt V2-inspired teacher network first extracts rich hierarchical representations using large-kernel depthwise convolutions, Layer Normalization, and Global Response Normalization (GRN). These representations guide the training of a lightweight MobileNetV2 student network through knowledge distillation.

During training, the student learns simultaneously from ground-truth labels and softened probability distributions produced by the teacher. This dual supervision allows the student to approximate the representational behavior of the teacher while maintaining substantially lower computational complexity.

4.1 Teacher Network: ConvNeXt V2-Inspired Architecture

The teacher network is a ConvNeXt V2-inspired convolutional architecture designed to capture high-level semantic features from plant leaf images. The design is motivated by the modernized convolutional architecture of ConvNeXt [5] and the GRN-based representation learning introduced in ConvNeXt V2 [6].

Given an input image:

$$x \in \mathbb{R}^{224 \times 224 \times 3}, \quad (4)$$

the teacher network produces logits:

$$z_t = T_\phi(x), \quad (5)$$

where ϕ represents the teacher parameters.

Each teacher block consists of a depthwise convolution with a large kernel, followed by normalization and channel-wise transformations.

Depthwise Convolution Depthwise convolution performs spatial filtering independently for each input channel, reducing the computational cost compared with a conventional dense convolution.

The operation can be represented conceptually as:

$$F'_{h,w,c} = \sum_{i,j} K_{i,j,c} F_{h+i,w+j,c}. \quad (6)$$

The use of a large kernel provides a larger receptive field and enables the teacher to capture broader spatial context within plant leaf images.

Layer Normalization Layer Normalization stabilizes feature distributions during training.

For feature vector h , the mean is:

$$\mu = \frac{1}{d} \sum_{i=1}^d h_i, \quad (7)$$

and the variance is:

$$\sigma^2 = \frac{1}{d} \sum_{i=1}^d (h_i - \mu)^2. \quad (8)$$

The normalized feature is:

$$LN(h_i) = \gamma \frac{h_i - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta. \quad (9)$$

Layer Normalization improves feature stability during training.

Global Response Normalization Global Response Normalization enhances global feature scaling and channel competition, following the general design principle introduced in ConvNeXt V2 [6].

The global response for channel c is:

$$g_c = \sqrt{\sum_{h,w} F_{h,w,c}^2}. \quad (10)$$

The normalized global response is:

$$\hat{g}_c = \frac{g_c}{\frac{1}{C} \sum_{c'} g_{c'}}. \quad (11)$$

The resulting feature representation is:

$$F'_{h,w,c} = \gamma_c F_{h,w,c} \hat{g}_c + \beta_c + F_{h,w,c}. \quad (12)$$

GRN encourages diversity among channel responses and improves the representation capability of the teacher network.

4.2 Student Network: MobileNetV2

The student network is based on MobileNetV2, which uses depthwise separable operations, inverted residual blocks, and linear bottlenecks to reduce computational complexity [8].

The student model produces logits:

$$z_s = S_\theta(x), \quad (13)$$

where θ denotes the student parameters.

Each MobileNetV2 bottleneck block consists of three principal stages.

Expansion Layer The expansion operation is:

$$F_{exp} = ReLU(W_e F). \quad (14)$$

Depthwise Convolution The depthwise convolution is:

$$F_{dw} = ReLU(K_{dw} * F_{exp}). \quad (15)$$

Projection Layer The projection operation is:

$$F_{proj} = W_p F_{dw}. \quad (16)$$

Residual connections are applied when the input and output dimensions match, following the MobileNetV2 design [8].

4.3 Knowledge Distillation Training

Knowledge distillation transfers information from the teacher to the student by combining hard-label supervision with teacher-generated soft targets [9].

The student learns from both ground-truth labels and the softened probability distribution generated by the teacher.

Soft Distillation Loss Teacher probabilities are calculated using temperature scaling:

$$p_t = \text{softmax}\left(\frac{z_t}{T}\right). \quad (17)$$

Student probabilities are:

$$p_s = \text{softmax}\left(\frac{z_s}{T}\right). \quad (18)$$

The distillation loss is calculated using Kullback–Leibler divergence:

$$\mathcal{L}_{KD} = T^2 \sum_i p_{t,i} \log\left(\frac{p_{t,i}}{p_{s,i}}\right). \quad (19)$$

Here, T denotes the temperature parameter. Temperature scaling produces softer probability distributions that expose the relative similarity between disease classes [9].

Hard Classification Loss Standard cross-entropy loss is used for supervision from the ground-truth labels:

$$\mathcal{L}_{CE} = - \sum_i y_i \log(\hat{y}_i). \quad (20)$$

Total Training Loss The combined training objective is:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{CE} + (1 - \alpha) \mathcal{L}_{KD}. \quad (21)$$

The student parameters are optimized using the AdamW optimizer.

During distillation, the teacher network remains frozen while gradients are propagated through the student network.

4.4 Parameter Comparison and Model Efficiency

The ConvNeXt V2-inspired teacher model contains approximately 27.82 million parameters, while the MobileNetV2 student network contains approximately 2.23 million parameters.

The teacher therefore contains approximately $12.5\times$ more parameters than the student.

Table 2. Model Parameter Comparison

Model	Total Parameters (M)	Trainable Parameters (M)
ConvNeXt V2-Inspired Teacher	27.82	0
MobileNetV2 Student	2.23	2.23
Hybrid KD MobileNetV2	2.23	2.23

The parameter reduction between the teacher and student is:

$$\Delta P = 27.82 - 2.23 = 25.59 \text{ M.} \quad (22)$$

The percentage reduction is:

$$R_P = \frac{27.82 - 2.23}{27.82} \times 100 \approx 92\%. \quad (23)$$

4.5 Inference and Edge Deployment

After training, the teacher network is discarded and only the MobileNetV2 student model is deployed.

Predicted probabilities are computed as:

$$\hat{y} = \textit{softmax}(z_s). \quad (24)$$

The predicted disease class is:

$$\hat{c} = \arg \max_c \hat{y}_c. \quad (25)$$

This deployment strategy eliminates the computational cost of the teacher during inference and enables efficient plant disease classification on resource-constrained devices.

5 Experiments and Results

This section evaluates the proposed hybrid knowledge distillation framework for plant disease detection. We first compare multiple transfer learning architectures to establish the accuracy-efficiency trade-off. We then evaluate the proposed teacher-student framework.

5.1 Transfer Learning Model Comparison

Before constructing the knowledge distillation framework, we conducted a comparative study of several widely used transfer learning architectures, including ResNet50, MobileNetV2, EfficientNet-B0, DenseNet121, ConvNeXt-Tiny, and VGG16.

Table 3. Transfer Learning Model Comparison for Plant Disease Classification

Model	GFLOPs	Params (M)	Accuracy	F1-score	Train Time (s)
ResNet50	8.26	23.51	0.9800	0.9800	1803.25
MobileNetV2	0.65	2.23	0.9800	0.9800	616.76
EfficientNet-B0	0.83	4.01	0.9800	0.9800	832.95
DenseNet121	5.79	6.96	0.9733	0.9735	1673.18
ConvNeXt-Tiny	8.93	27.81	0.9733	0.9733	731.89
VGG16	30.93	134.27	0.9667	0.9667	4953.27

Table 4. Hybrid Knowledge Distillation Model Comparison

Model	Accuracy	F1-score	Params (M)	GFLOPs
MobileNetV2 Baseline	0.9800	0.9798	2.23	0.31
ConvNeXt V2-Inspired Teacher	0.9733	0.9733	27.82	4.47
Hybrid KD MobileNetV2	0.9800	0.9798	2.23	0.31

These architectures represent different approaches to balancing model depth, feature reuse, computational complexity, and parameter efficiency [1–8].

The models were evaluated using classification accuracy, F1-score, computational complexity (GFLOPs), number of parameters, and training time.

The results are summarized in Table 3.

MobileNetV2 achieves a strong accuracy-efficiency trade-off, obtaining 0.9800 accuracy and 0.9800 F1-score while requiring only 2.23 million parameters.

In contrast, ConvNeXt-Tiny contains approximately 27.81 million parameters. Its greater capacity motivates its use as a high-capacity teacher, whereas MobileNetV2 is selected as the lightweight student.

5.2 Hybrid Knowledge Distillation Model Evaluation

After applying knowledge distillation, we evaluated the performance of the teacher model, baseline MobileNetV2, and distilled MobileNetV2.

The results are presented in Table 4.

The distilled MobileNetV2 student achieves the same reported classification performance as the baseline MobileNetV2 while maintaining the same compact deployment architecture.

The results therefore demonstrate performance preservation rather than an accuracy increase. Knowledge distillation provides training supervision from the higher-capacity teacher without increasing the parameter count of the final student model.

5.3 Efficiency Analysis

The proposed knowledge distillation approach substantially reduces model complexity relative to the high-capacity teacher while preserving high predictive performance.

The parameter reduction between the teacher and student models is:

$$\Delta P = 27.82 \text{ M} - 2.23 \text{ M} = 25.59 \text{ M}. \quad (26)$$

The corresponding parameter reduction is approximately:

$$\frac{25.59}{27.82} \times 100 \approx 92\%. \quad (27)$$

Using the reported final student computational cost of 0.31 GFLOPs, the computational difference relative to the teacher is:

$$\Delta F = 4.47 - 0.31 = 4.16 \text{ GFLOPs}. \quad (28)$$

The relative computational reduction is:

$$R_F = \frac{4.47 - 0.31}{4.47} \times 100 \approx 93.1\%. \quad (29)$$

The reduction in GFLOPs and parameter count demonstrates the computational advantage of deploying the lightweight student rather than the high-capacity teacher.

5.4 Generative AI Integration

Following classification, the predicted disease label and confidence score are provided to a Generative AI module implemented using Ollama.

The module generates contextual agricultural guidance including:

- Confirmation of plant health status.
- Preventive care steps.
- Monitoring recommendations.
- Irrigation-related guidance.
- Fertilizer recommendations.
- General pest and disease management suggestions.
- Recommended follow-up monitoring.

The integration of this advisory component extends the system beyond conventional classification by converting the predicted disease category into contextual agricultural guidance.

6 Discussion

The experimental results demonstrate the importance of considering both predictive performance and computational efficiency when designing plant disease detection systems.

Although high-capacity architectures can learn rich visual representations, their parameter and computational requirements can make direct deployment difficult. The proposed teacher–student strategy separates the requirements of model training and deployment.

During training, the ConvNeXt V2-inspired teacher provides high-capacity representations and soft class probabilities. During deployment, the teacher is removed and only the lightweight MobileNetV2 student remains.

This design provides two primary advantages. First, the student retains a compact architecture suitable for edge devices. Second, the student receives additional supervision from the teacher beyond the information contained in one-hot ground-truth labels.

The experimental results show that the final student maintains the same reported accuracy and F1-score as the baseline MobileNetV2 while requiring substantially fewer parameters than the teacher.

The parameter reduction from 27.82 million to 2.23 million parameters corresponds to approximately 92% reduction. The reported computational comparison also indicates a substantial reduction in GFLOPs when the lightweight student is deployed.

The integration of Generative AI further extends the practical functionality of the framework. The classifier identifies the likely disease, while the advisory module converts the prediction into understandable agricultural guidance.

However, the current evaluation is based on a relatively small dataset configuration. Therefore, larger and more diverse agricultural datasets should be considered in future experiments to evaluate robustness under additional environmental conditions.

7 Conclusion

This work presents an edge-efficient hybrid knowledge distillation framework for plant disease detection. The proposed architecture combines a ConvNeXt V2-inspired teacher with a lightweight MobileNetV2 student.

The teacher uses large-kernel depthwise convolution, Layer Normalization, and Global Response Normalization to learn rich visual representations. These representations are transferred to MobileNetV2 through a dual-loss knowledge distillation objective combining hard-label cross-entropy and temperature-scaled Kullback–Leibler divergence.

Experimental comparison across multiple transfer learning architectures demonstrates that MobileNetV2 provides a favorable accuracy-to-computation trade-off. The final student contains only 2.23 million parameters compared with 27.82

million parameters in the teacher, resulting in approximately 92% parameter reduction.

The reported computational comparison further indicates a substantial reduction in GFLOPs when the lightweight student is deployed instead of the high-capacity teacher.

The final student maintains the reported classification performance of the baseline MobileNetV2 while benefiting from teacher supervision during training. Thus, the principal advantage of the proposed framework is computationally efficient deployment rather than an increase in classification accuracy.

An additional Generative AI advisory module implemented using Ollama transforms the classification output into contextual agricultural recommendations. This provides a pathway from disease recognition toward practical agricultural decision support.

Overall, the proposed hybrid knowledge distillation framework demonstrates that high-capacity convolutional architectures can provide training supervision to lightweight networks, enabling computationally efficient plant disease detection with potential for resource-constrained agricultural edge applications.

Future work may explore extending the framework to larger agricultural datasets, incorporating temporal crop monitoring using sequential imagery, optimizing the model for specific edge hardware, and extending the Generative AI component to multilingual and region-specific agricultural recommendations.

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